A Child's Right to Play: Results from the Brain-Computer Interface Game Jam 2019 (Calgary Competition)

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Abstract— **Children with severe neurological disabilities may be unable to communicate or interact with their environments, depriving them of their right to play. Brain-computer interfaces (BCI) offer a means for such children to control external devices using only their brain signals, thereby introducing new opportunities for interaction. We organized the first North American BCI Game Jam to incite the development of BCI-compatible games for children. Nine games were submitted by 30 participants across North America. Games were judged by researchers and disabled children currently using BCI. Preliminary results demonstrate variety in game criteria preferences amongst the children who judged the games. The BCI Game Jam demonstrated promising potential for the creation of enjoyable games to suit the individual needs and preferences of children with severe neurological disabilities.**

Clinical Relevance—**This paper highlights a potential strategy for creating a variety of BCI-compatible video games for a diverse group of pediatric patients with severe neurological disabilities that suit their individual preferences.**

I. Introduction

Thousands of Canadian children are living with profound physical disabilities, rendering them unable to walk, talk, or use their hands[1]. Children with such severe disabilities are fully dependent on others for all activities of daily living and deprived of their fundamental rights including communication and recreation [2]. Even more unfortunate is that a portion of these children are intellectually normal[2]. With virtually no way to communicate or interact with their environments, such cognitively aware and highly capable children are literally trapped inside a body that does not work.

Brain-computer interfaces (BCI) are an emerging technology that represent a promising opportunity for such children to interact with their environments[3]. BCI can function by non-invasively acquiring, analyzing and translating brain signals into commands using electroencephalography (EEG). The translated commands are then used to control an external device, thereby bypassing severe neuromuscular limitations[4]. Such mental control signals can be produced in response to external stimuli (evoked potentials and P300 event-related potentials) or they can be generated internally by imagined movements (motor imagery)[5],[6]. BCIs are entirely safe, painless, and adaptable to home-based use. While there has been a dramatic increase in recent BCI development, the majority of studies have been in adults, resulting in neglect of pediatric populations and their specific needs[6].

Engaging in play is recognized as a child's right by the United Nations General Assembly[7]. Play is important for promoting healthy child development, and can come in a variety of forms, including video game play[8]. A 2018 report by the Entertainment Software Association of Canada revealed that over 90% of Canadian children and adolescents play video games[9]. Recent research has demonstrated that video game play has important benefits for social, cognitive, and emotional development[10]. Importantly, developmental psychologists have suggested that engaging in this type of play has the potential to be largely beneficial for enhancing mental health and well-being in children[10].

Children with physical disabilities have many of the same interests as their typically developing peers, including playing video games. However, children with motor impairments encounter barriers in both the hardware (controllers), which assumes bimanual hand coordination, as well as with the software, which assumes that the user can access and use all functions of the controllers[11]. Unfortunately, many commercial gaming systems have not been designed with such individuals in mind.

In recent years, large technology companies have teamed up to develop adaptive controllers for people with physical disabilities. Microsoft partnered with groups such as The AbleGamers Foundation and The Cerebral Palsy Foundation to build the Xbox Adaptive Controller for improved accessibility in video game control[12]. After its release in September 2018, Logitech G collaborated with the Microsoft team to create the Adaptive Gaming Kit, which contains an assortment of buttons and switches compatible with the Xbox Adaptive Controller[13].These adaptive gaming technologies were designed to support inclusivity in gaming; however, the challenge of inclusivity remains unaddressed for children that have no functional movement at all.

We therefore organized the first North American BCI Game Jam over the weekend of November 8-10th, 2019 . The event aimed to attract game developers to create child-friendly BCI-compatible games suitable for the needs of children with severe neurological disabilities. Game jams are an accelerated game development competition focusing on specific themes or constraints. Participation in game jams has

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significantly grown in the past decade, with events like Ludum Dare growing from 120 game submissions in 2009 to over 2500 submissions in 2019[14]. With the expanding community, game jams have become a fundamental platform for prototyping innovative game designs and ideas, all while garnering critical feedback from a widespread audience. Game jam competitions thus represent a unique ecosystem where developers push creative boundaries in game design using a platform that provides potential market-testing and prototype feedback.

This paper describes the details from the BCI Game Jam 2019. Judging for the BCI Game Jam is pending completion. The winner will be determined by the unique feedback contributed by end-users (children with severe physical disabilities). In addition, feedback will help improve future pediatric BCI application development and provide game developers constructive criticism about which aspects of their BCI games are enjoyable to children and what could be improved, thus translating BCI theory to application.

II. Methods

A. BCI Game Jam Participant Recruitment

Participants were recruited to join the BCI Game Jam 2019 at one of two in person sites (hosted in Calgary and Edmonton, Alberta, Canada) or remotely through the itch.io webpage (https://itch.io/jam/bci-game-jam). Recruitment was amplified through various outreach methods. For the Calgary and Edmonton local sites, the event was promoted through university clubs, hospital organizations and promotional flyers. To reach the wider community, remote game development/engineering/ innovation communities were contacted using channels on popular communication platforms including Discord, Slack and itch.io. Participants were invited to complete a voluntary demographics survey. To help introduce participants to the BCI technology, simple videos covering BCI topics were recorded and uploaded on YouTube (Neural Matrix Entertainment).

B. Judging Participants

Four children with severe quadriplegic cerebral palsy (no functional movement, 1 female, average age 10.25) currently enrolled in the Pediatric BCI Program at the Alberta Children's Hospital (ACH) were verbally invited, along with their parents and caregivers, to participate as judges of the 2019 BCI Game Jam event at the Calgary Site. Families attended a pre-event (Game Jam Kick-Off) where there were open interactions between the game developers and families. Participants were able to ask children and parents about their interests, likes and dislikes, and future potential applications that might come from the game development.

C. Design Guidelines and Equipment

Game Jam participants were allotted 48 hours to design BCI-compatible games to be played using either motor imagery or the visual P300 response. For motor imagery-based games, we used the Emotiv EPOC+ (Emotiv Systems Inc., USA). For P300-based games, a g.tec BCI system (Guger Technologies, Austria) was used. The P300 games were developed and tested using the P300 Dynamic Cube (https://github.com/ekinney-lang/bci-game-jam2019.).

D. Study Procedures

This observational study aimed to generate feedback on the contributions submitted to the BCI Game Jam competition for the purposes of determining the outcome of the event.

Game submissions were judged by researchers at the University of Calgary (n=5), who completed a standardized Judging Survey to evaluate 3 categories associated with the game: rule clarity, implementation of BCI principles, and game play. Each submission was scored out of 45 points, with 5 possible bonus points for including interactive elements.

Prior to evaluating the games, the child judges, with the assistance of their parents or caregivers, were asked to complete a Pre-Judging Ranking Survey. They were asked to rank six game criteria in order of least important (1) to most important (6): Theme, Art/Graphics, Music/Sound, Challenge (difficulty within the game), Enjoyment, and Replayability.

Information from the Pre-Judging Ranking Survey was used to linearly weight questions of the game Judging Survey for each individual child judge, with the most important criteria receiving 6 points, the next 5 points, etc. This allowed for a more personalized evaluation of the games from each individual judge. Participants then had the opportunity to play each game submission (n=9, as available) during individual 1-hour long sessions. Games were played for as long as the participants were engaged and interested. Following each game, they were asked a series of six yes/no questions to their voluntary opinions about what they liked/disliked about the games. These surveys were scored out of 21 points.

Final results are pending judging completion. The final scores for each submission will be calculated out of 10, with scores weighted 3/7 for the research judges/child judges. The scores from the Judging Survey for Researchers (45 points) will be divided by 15 to get a score out of 3. The scores from the Judging Survey for Children (21 points) will be divided by 3 to get a score out of 7. The final scores will be calculated as the sum of the adjusted scores for each Judging Survey.

III. Results

A. BCI Game Jam Participants

Forty-six people registered as participants for the BCI Game Jam in Calgary (n=25) and Edmonton (n=21). Thirteen participants from Calgary attended the Game Jam Kick-Off event, which included an introduction to BCIs, a panel discussion from the Pediatric BCI Research Team at ACH, and a panel discussion from three families currently enrolled in the BCI Program at ACH. Seven participants responded to a demographics survey (Table. 1). Submissions were made by additional participants across North America including Calgary, Edmonton, Toronto, Illinois and Florida.

B. Game Submissions

A total of nine games were submitted at the end of the 48-hour game jam. All games were made using the Unity game engine (Unity Technologies). Six of those games were motor imagery-based and three were P300-based. One motor imagery-based submission was a 3D interactive game which

TABLE 1. DEMOGRAPHICS OF THE BCI GAME JAM PARTICIPANTS

Question	**Responses**
Age	26, 25, 19, 21, 25, 30, 18
Gender	6 male, 1 female
Highest educational attainment	42.9% High school diploma; 14.3% Post-secondary diploma; 28.6% Bachelor's degree; 14.3% Master's degree
Educational background	42.9% Engineering; 28.6% Computer science; 14.3% Math; 14.3% Hard sciences
Game development experience	100% Less than a year
Number of game jams attended	100% 1 (including this one)
Number of people on team	42.9% Three; 14.3% One; 14.3% Four; 14.3% Six; 14.3% Seven; 14.3% Eight
Role in the team	100% Programming; 57.1% Design; 14.3% Healthcare Advisor; 14.3% Art
Role outside of the game jam	85.7% Student – related to (non-game) software development; 14.3% Software developer

involved navigating a specialized robot through a maze. As this game was not accessible to test outside of its participants' location, it was ineligible to be included in the game judging.

A simple summary of the submitted games are provided below. Interested parties wanting access to these submissions should contact the corresponding author (DK).

Motor Imagery-based Games

- Endless Runner: Jump over obstacles as you run across a mountain range. (Theme: Obstacle avoidance, Single Player)
- Yummy Yucky: Build a story by choosing different characters and their favourite/least-liked foods. (Theme: Story, Single Player)
- Space Brainz 2: Choose a path and battle your way to fight the Big Brain Spacecraft. Choose from several types of weapons on your ship to help you in battle. (Theme: Space, Single Player)
- Zap Attack: Shoot ducks as they enter your target area. Get more points for catching them in the bullseye. (Theme: Hunting, Single Player)
- Sumo Bootle: Slam into your opponents around the arena to weaken them. Hit 'health' boxes to regain your health (Fig. 1a). (Theme: Battle, Multiplayer)

P300-based Games

- Kolor-Kaze: Paint on a virtual canvas by launching a ball around an arena which changes colors based on the last-hit wall. (Theme: Art, Single Player)
- Planet Defender: Shoot down the enemy spacecrafts before they invade your planet (Fig. 1b). (Theme: Action, Single Player)
- Kirby Pop: Launch your character into the obstacles to knock them down and eat cake. (Theme: Puzzle/Strategy, Single Player)

C. Judging Assessments from Researchers

Five researchers from the Pediatric BCI Team at ACH completed Judging Surveys for eight of the nine submissions. The top three highest scored games were: Sumo Bootle, Planet Defender, and Kirby Pop.

D. Judging Assessments from Children

Three of the four invited children from the Pediatric BCI Program at ACH have participated as judges of the Game Jam at this time. They ranked the most important game criteria in the Pre-Judging Survey as Theme, Music/Sound, and Enjoyment (Table 2). All three children played and judged the 5 motor imagery-based games. Preliminary results show that the top three highest scored games by child judges were Sumo Bootle, Zap Attack, and Yummy Yucky (Table 3).

IV. DISCUSSION

Preliminary findings from the BCI Game Jam 2019 are promising. Results from the Participant Demographic Survey indicate a variety in age, educational attainment and educational background of participants. It is estimated that there were approximately 30 participants, resulting in a

A) B)

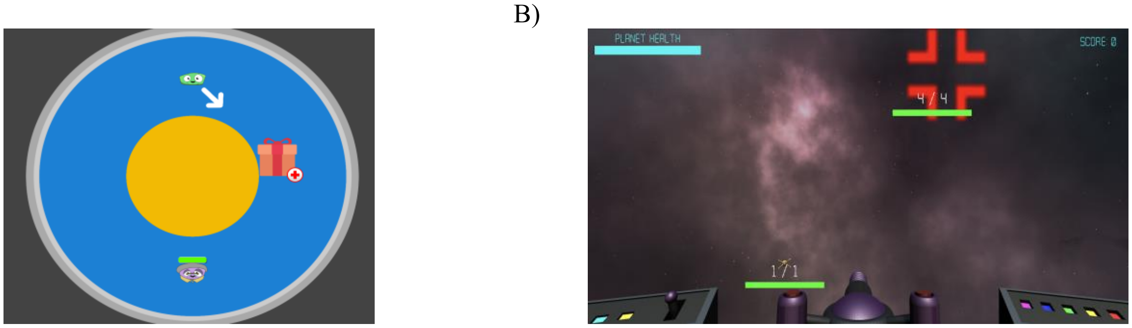

Figure 1. Game submissions. A) Sumo Bootle motor-imagery based game. The arrow rotates around the sumo character and the goal is to aim in the direction of your opponent to knock down their health or aim toward the 'health' box to power up your health levels. B) Planet Defender P300-based game. The reticle flashes randomly in front of the player's spacecraft. The goal is to attend to the invading spacecraft and silently count the number of times that the reticle flashes on the target spacecraft. If the invading spacecrafts get past to your planet, health is lost.

TABLE 2. RESULTS OF THE PRE-JUDGING SURVEY

Game Criteria	Judge 1	Judge 2	Judge 3
Theme	3	6	3
Art/Graphics	4	4	4
Music/Sound	6	2	2
Challenge	2	3	1
Enjoyment	5	1	6
Replayability	1	5	5

TABLE 3. WEIGHTED SCORES FROM THE PRE-JUDGING AND JUDGING SURVEY FOR CHILDREN

Game	Judge 1	Judge 2	Judge 3	Average
Endless Runner	13.9	3.4	18	11.8
Yummy Yucky	19.3	8.1	16.9	14.8
Space Brainz 2	17	2	15.8	8.9
Zap Attack	15.2	11.9	20.7	15.9
Sumo Bootle	19.3	21	17.8	19.4

response rate of 23%. This rate is much higher than what has been found in surveys from the Global Game Jam, which is likely due to the relatively small number of participants in the current study. As participation in the Global Game Jam has increased, the corresponding response rate has decreased[15]. Respondents were all young adults, ranging in age from 18 to 30 with the highest educational attainment among the majority of respondents being a high school diploma. As most respondents indicated they were students, it is likely that most participants were undergraduate students. This is consistent with demographics results from the Global Game Jam [15]. All of the respondents had less than a year of game development experience and indicated that this was their first game jam, implying that enthusiasm for the event was high.

The variety of game types and themes submitted suggest positive outcomes of the BCI Game Jam in stimulating a multitude of game ideas for a diverse group of judges.

The heterogeneity in the results of the Pre-Judging Survey demonstrate the unique preferences of children when playing video games. This is consistent with other studies that have shown that children's game preferences vary across age and gender[16], [17]. However, it is important to note that all three judges rated 'Art/Graphics' as fairly important, at a 4 out of 6. In comparing the responses of the Judging Survey for Children, it suggests that a child's perceived enjoyment of a video game may be positively correlated with how much they liked the graphics.

Correspondingly, the games were scored very differently by each judge. For Judge 1, a 12-year-old boy, 'Yummy Yucky' and 'Sumo Bootle' tied for top game, with 'Space Brainz 2' in second place. For Judge 2, a 13-year-old girl, 'Sumo Bootle' received a perfect score, while she gave 'Space Brainz 2' the lowest score in the competition. Judge 3, a 6-year-old boy, scored 'Zap Attack' as his top game.

V. CONCLUSION

The aim of this paper was to disseminate findings and critical feedback from both the participants and judges of the BCI Game Jam 2019. Preliminary results demonstrate the promising potential for game jams to stimulate the creation of BCI-compatible games for children with severe motor impairments that suit their individual preferences. Future work may evaluate the use of personal preference for game elements to optimize performance on utility-driven applications, such as spelling grids or cursor control.

REFERENCES

[1] P. Minocha, S. Sitaraman, and P. Sachdeva, "Clinical Spectrum, Comorbidities, and Risk Factor Profile of Cerebral Palsy Children: A Prospective Study," *J. Pediatr. Neurosci.*, vol. 12, no. 1, pp. 15–18, 2017, doi: 10.4103/1817-1745.205622.

[2] M.-A. Bruno *et al.*, "Locked-In Syndrome in Children: Report of Five Cases and Review of the Literature," *Pediatr. Neurol.*, vol. 41, no. 4, pp. 237–246, Oct. 2009, doi: 10.1016/j.pediatrneurol.2009.05.001.

[3] J. N. Mak and J. R. Wolpaw, "Clinical Applications of Brain-Computer Interfaces: Current State and Future Prospects," *IEEE Rev. Biomed. Eng.*, vol. 2, pp. 187–199, 2009, doi: 10.1109/RBME.2009.2035356.

[4] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, "Brain-computer interfaces for communication and control," *Clin. Neurophysiol. Off. J. Int. Fed. Clin. Neurophysiol.*, vol. 113, no. 6, pp. 767–791, Jun. 2002.

[5] R. A. Ramadan and A. V. Vasilakos, "Brain computer interface: control signals review," *Neurocomputing*, vol. 223, pp. 26–44, Feb. 2017, doi: 10.1016/j.neucom.2016.10.024.

[6] U. Chaudhary, N. Birbaumer, and A. Ramos-Murguialday, "Brain-computer interfaces for communication and rehabilitation," *Nat. Rev. Neurol.*, vol. 12, no. 9, pp. 513–525, Sep. 2016, doi: 10.1038/nrneurol.2016.113.

[7] United Nations General Assembly, "Convention on the Rights of the Child," New York, Nov. 1989.

[8] K. R. Ginsburg, "The Importance of Play in Promoting Healthy Child Development and Maintaining Strong Parent-Child Bonds," *Pediatrics*, vol. 119, no. 1, pp. 182–191, Jan. 2007, doi: 10.1542/peds.2006-2697.

[9] Entertainment Software Association of Canada, "Essential Facts About The Canadian Video Game Industry," 2018.

[10] I. Granic, A. Lobel, and R. C. M. E. Engels, "The benefits of playing video games.," *Am. Psychol.*, vol. 69, no. 1, pp. 66–78, 2014, doi: 10.1037/a0034857.

[11] D. Hobbs, M. Henschke, B. Wilkinson, and K. Reynolds, "Game on! Accessible gaming for children with disabilities," presented at the National Conference of the Australian Rehabilitation & Assistive Technology Association, Sydney, Australia, 2012.

[12] J. Ore, "Xbox's Adaptive Controller aims to bring gaming to community of disabled players," CBC Radio, 03-Sep-2019.

[13] "Logitech G Adaptive Gaming Kit Creates New Possibilities for Gamers with Accessibility Needs," *BusinessWire*, 18-Nov-2019.

[14] "Ludum Dare," *Wikipedia*. [Online]. Available: https://en.wikipedia.org/wiki/Ludum_Dare.

[15] T. Steinke, M. Linsenbard, E. Fiske, and F. Khosmood, "Understanding a Community: Observations from the Global Game Jam Survey Data," in *Proceedings of the International Conference on Game Jams, Hackathons, and Game Creation Events - GJH&GC '16*, San Francisco, CA, USA, 2016, pp. 15–21, doi: 10.1145/2897167.2897173.

[16] C. Quaiser-Pohl, C. Geiser, and W. Lehmann, "The relationship between computer-game preference, gender, and mental-rotation ability," *Personal. Individ. Differ.*, vol. 40, no. 3, pp. 609–619, 2006, doi: 10.1016/j.paid.2005.07.015.

[17] J. Fromme, "Computer Games as a Part of Children's Culture," *Game Stud.*, vol. 3, 2003.